Rural Legal Aid Summarizer: Literature Review, Comparative Analysis, and Implementation Status

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***Abstract*—Access to justice in India is constrained not only by the availability of law, but also by the difficulty ordinary citizens face in understanding legal language, procedures, and personal legal documents. This paper reviews academic work on legal document summarization and retrieval-augmented generation (RAG), examines government and private legal-information systems, and analyzes the position of the Rural Legal Aid Summarizer within this landscape. The reviewed project uses document-specific retrieval, multilingual text and audio output, OCR for scanned documents, and a free/open-source, low-resource deployment model. The analysis identifies a gap between general legal question-answering systems and tools that explain a citizen's own uploaded document in regional languages, highlights retrieval mismatch and legal hallucination as important evaluation concerns, and identifies summary-augmented chunking, quantitative retrieval evaluation, and an explicit "information not found in document" fallback as useful improvements. Building on this positioning, the paper further documents the project's current implementation status: the capabilities that have been built and are functioning, and the concrete gaps – most notably summary export and access to previous summaries – that remain on the development roadmap.**

***Index Terms*—Legal document summarization; Retrieval-Augmented Generation; RAG; legal NLP; access to justice; OCR; multilingual AI; rural India; ChromaDB; system implementation.**

I. Introduction

Access to justice in India is constrained less by the absence of law and more by the inability of ordinary citizens – especially in rural areas – to understand legal language, procedures, and their own documents. The Rural Legal Aid Summarizer is positioned at the intersection of legal document summarization, retrieval-augmented generation (RAG), and access-to-justice technologies.

This paper considers two bodies of work: academic research on legal document summarization and RAG, and deployed government and private systems that address parts of the same problem. It then presents a gap analysis describing how the project's design choices differentiate it from the surveyed alternatives, followed by a documented account of what has actually been implemented at the current development stage and what remains outstanding.

II. Academic Literature Review

A. Legal Document Summarization: Extractive vs. Abstractive

Survey work classifies legal summarization along extractive versus abstractive and supervised versus unsupervised approaches. One representative method assigns rhetorical roles such as facts, arguments, ratio, and conclusion to judgment sentences and composes a summary from high-value roles. A 2026 short review spanning approximately 150 sources from 2004–2025 reports a recurring concern: automatic summaries may perform well on overlap metrics such as ROUGE while still containing legally significant inaccuracies [2].

This observation directly motivates a verification-oriented architecture. The project grounds generated statements in retrieved source text and is designed to return "information not found" rather than allowing unsupported content to be filled by plausible-sounding generation.

B. Retrieval-Augmented Generation in the Legal Domain

RAG combines a retriever that surfaces relevant passages with a generator that composes an answer conditioned on the retrieved context. Surveys identify summarization, dialogue, and specialized domains such as law and medicine as important RAG applications [4].

A 2025 legal-NLP survey observes that, although legal document summarization is an increasingly common LLM use case, RAG integration into legal document summarization remains comparatively underexplored relative to legal question-answering and legal judgment prediction. RAG-grounded summaries are intended to improve factual consistency by tying generation to relevant statutes and case references.

Domain-adapted approaches have also addressed hallucination-free legal summarization through hybrid retrieval. One framework combines BM25 with a legal-aware summarization model and reports that top-k chunk selection reduces redundant or irrelevant content passed to the generator [5]. This mirrors the project's hybrid dense-embedding and sparse-BM25 retrieval design.

LegalBench-RAG highlights retrieval precision and recall and identifies document-level retrieval mismatch as a failure mode of naive chunking. Summary Augmented Chunking (SAC), which attaches a short document-level summary to each chunk before embedding, is reported as a method for reducing this mismatch [7].

C. Access-to-Justice-Specific Systems in India

LawPal (2025) is a RAG-based system explicitly framed around enhanced legal accessibility in India and addresses citizens who cannot easily parse formal legal English [3]. A

2024 Legal Guide chatbot proposal for marginalized Indian communities combines a multilingual legal-information repository, a locally run LLM, and a legal-aid directory, but focuses on general legal question-answering rather than document-specific summarization [10].

A 2026 multilingual legal document analyzer reports approximately 92% clause-extraction precision and 88% question-answering accuracy for a transformer-based system combining clause extraction, RAG-based Q&A, abstractive summarization, and translation across several Indian languages [9]. These figures provide a useful reference point for quantitative evaluation of systems of this type.

D. Key Takeaways

1) Grounding generation in retrieved source text is a central mitigation strategy for hallucination in legal summarization.
2) Chunk-level retrieval mismatch is a documented failure mode with published mitigation strategies.
3) Overlap-based evaluation metrics are insufficient by themselves for legal correctness, motivating verification and explicit fallback behavior.
4) India-specific prototypes validate the access problem but are generally narrower than a multilingual, audio-enabled, document-specific summarizer designed for low-resource deployment.

III. Existing Products

A. Government and Public-Sector Systems

Table I summarizes reviewed government and public-sector systems.

TABLE I
GOVERNMENT / PUBLIC-SECTOR SYSTEMS

System	What it does	Relevant limitation
NALSA Legal Services app	Legal-aid applications, tracking, contact with Legal Services Authorities	Administrative/case-tracking tool; does not read, summarize, or explain uploaded documents [18]
NALSA Legal Services Mgmt. System	Bilingual e-portal for applications, document upload, tracking, reminders	Document upload is for submission to an authority, not AI explanation to the citizen [17]
VANI (NIC)	Multilingual conversational-AI framework with translation and voice	Institutional infrastructure, not an open personal document-summarization assistant [19]
NyayMarg (UPSLSA, 2025)	Hindi chatbot answering ∼1,500 pre-compiled legal Q&A pairs	Fixed Q&A bank; cannot explain a citizen's specific uploaded notice or agreement [20]
Jugalbandi	Multilingual generative-AI assistant for government-aid info and rights awareness	General information assistant, not focused on an individual's uploaded legal document [21]

B. Private and Commercial Products in India

Table II summarizes reviewed private and commercial systems.

TABLE II
PRIVATE / COMMERCIAL SYSTEMS

Product	Core offering	Gap vs. project
Legal Document Summarizer	Gemini-based summarization, OCR for scans, follow-up chatbot	English-only; no stated Hindi/Marathi audio; requires users to supply a Gemini API key [11]
LegalEase AI	Contract/NDA/rental analysis and risky-clause detection in plain English	English-only; paid features; no stated regional-language/audio pipeline [12]
Draft Bot Pro / Law Bot Pro	Case summarization, drafting, research, citations, PDF Q&A	Primarily for lawyers/students; not positioned for rural lay users [13]
AI Lawyer: Indian Law Expert	General Indian-law Q&A, drafting, case analysis, document review	General chatbot rather than document-grounded RAG; no regional-language audio [14]
LawCentral AI	Case-law research, judgment summarization, statute search, document tools	Research workflow oriented; not primarily a plain-language personal-document explainer [15]
LegalMitra	Hindi-first legal chat, document generation, case tracking; markets toward rural India	Closest positioning, but paid limits and emphasis on generation rather than explaining an existing document [16]

C. International Reference Points

Large commercial legal-AI platforms such as Thomson Reuters' CoCounsel-style tools, Casetext, Lawgeex, Rocket Lawyer, and DoNotPay apply retrieval and grounded generation to legal research, contract review, consumer automation, or related workflows. These systems are oriented toward Western legal systems, English-language use, and commercial workflows, making them poor direct matches for a rural Indian citizen seeking an explanation of a personal regional-language document.

IV. Gap Analysis

Across the academic prototypes and deployed products surveyed, the project occupies a combination of capabilities that is not represented by a single reviewed system.

1) **Document-specific grounding:** many citizen-facing tools answer from general knowledge bases or fixed Q&A banks, while the project is designed around RAG over the user's own uploaded document.
2) **Regional-language text and audio together:** the reviewed project combines OCR in English/Hindi/Marathi with multilingual summary and gTTS audio output, while alternatives typically provide only part of this combination.
3) **Free, open-source, low-resource deployment:** the project is designed around free/open-source components and modest laptop-class hardware without a GPU, in

contrast to commercial products that may cap free use or gate features behind paid tiers.

4) **Explicit "not found in document" grounding:** the project treats refusal to invent unsupported information as a design principle rather than only a user-interface feature.

V. Comparative Positioning

Table III summarizes how the project compares to the closest reviewed alternatives along key dimensions.

TABLE III
Comparative Positioning

Dimension	Rural Legal Aid Summarizer	Closest reviewed alternatives
Own-document RAG	Yes; hybrid dense + BM25 retrieval over ChromaDB	LawPal; Draft Bot Pro
Regional output	English, Hindi, Marathi; text + audio	LegalMitra: Hindi text; VANI: voice but not document-specific
Scanned documents	OCR using PyMuPDF/OpenCV/PyTesseract; 3 languages	Legal Document Summarizer: OCR, English-only
End-user cost	Free/open-source stack	Many private tools cap free use or paywall features
Target user	Rural citizens with low legal/digital literacy	Mostly lawyers, students, or urban consumers
Hallucination mitigation	RAG + explicit fallback + retry/backoff + caching	Rarely documented publicly in commercial tools

VI. System Overview

The system is designed to run entirely on free and open-source components, on a single modest development machine (Intel i5-10210U, 8 GB RAM, no dedicated GPU), reflecting the low-resource deployment constraint identified above. The architecture consists of a React (Vite, Tailwind, Axios) frontend and a Python Flask backend, communicating over a stateless HTTP API. The backend performs four sequential stages on each uploaded document: optical character recognition, hybrid retrieval against a curated legal knowledge base, grounded summarization via a large language model, and multilingual text-to-speech synthesis.

VII. Completed Capabilities

The following components have been implemented and are functional in the current build.

A. Document Ingestion and OCR

The system accepts scanned or photographed documents and extracts text using PyMuPDF for born-digital PDFs and a PyTesseract/OpenCV pipeline for scanned or image-based input, with trained OCR support for English, Hindi, and Marathi (`eng+hin+mar`). This directly addresses a gap noted in the comparative review, where most competing tools either lack OCR entirely or support English only.

B. Curated Legal Knowledge Base

Retrieval is grounded in a structured knowledge base organized into five domains: acts, banking, circulars, court notices, and land records. This scopes the system to the categories of documents rural citizens most commonly encounter, rather than attempting general-purpose legal coverage.

C. Hybrid Retrieval

The retrieval layer combines dense embeddings, generated with the `fastembed` library using the `BAAI/bge-small-en-v1.5` model, indexed in ChromaDB, with a sparse BM25 index (`rank_bm25`). This hybrid dense-sparse design mirrors approaches identified in the literature review as effective for reducing irrelevant or redundant retrieved content before generation.

D. Grounded Summarization

Summaries are generated using the Google Gemini API (`gemini-3.5-flash-lite / gemini-3.5-flash`), conditioned on the retrieved passages rather than on the model's parametric knowledge alone. This retrieval-grounded design is intended to mitigate the hallucination risk repeatedly flagged in the legal-summarization literature, where high overlap-based scores do not guarantee legal correctness.

E. Multilingual Text and Audio Output

Summaries are produced in English, Hindi, and Marathi, with an audio rendering of the summary generated via gTTS. This combination of OCR, summary, and audio across three languages is, per the comparative analysis above, not offered together by any single reviewed competitor.

F. Lightweight Session Handling

The backend uses SQLite in write-ahead-logging (WAL) mode with a generated session identifier per visit. No user account, login, or authentication is required, removing a friction point identified as a barrier for first-time or low-digital-literacy users.

G. Frontend and Interface

The frontend is built in React (Vite) with Tailwind CSS and Axios for API communication. An interface redesign is in progress, moving from the current dark navy-and-orange theme toward a lighter default theme with a dark-mode toggle, using Stitch (Google's AI-assisted UI design tool) to accelerate layout iteration.

VIII. Pending Work

The following capabilities are identified as outstanding at the current stage of development, in order of priority.

A. Summary Export / Download

The system does not yet allow a user to download a generated summary (e.g., as a PDF or plain-text file) for offline use, printing, or sharing with a family member or local official. Given the target user's limited and often intermittent connectivity, an offline-retainable copy of the summary is a practical necessity rather than a convenience feature.

B. Access to Previous Summaries

The system does not currently expose a way for a user to view summaries generated earlier in the same session, or across sessions. Because the system is session-based rather than account-based, this requires a design decision: whether to persist a short-lived, locally-scoped history keyed to the existing session identifier (preserving the no-login design principle) or to introduce optional lightweight accounts. The former better preserves the project's low-friction access model.

C. Quantitative Retrieval and Summary Evaluation

The literature review noted that reviewed academic and commercial systems rarely publish retrieval precision/recall or summary-quality metrics, and recommended that this project do so rather than relying only on qualitative examples. This evaluation has not yet been conducted and remains a priority for validating the retrieval and summarization pipeline against a labeled sample of documents.

D. Explicit Fallback Documentation

The system is designed to avoid generating unsupported content when the knowledge base does not contain relevant information, but this "information not found in document" behavior has not yet been formally documented or systematically tested as a design principle.

E. Summary-Augmented Chunking

Adoption of Summary-Augmented Chunking (SAC), in which a short document-level summary is attached to each retrieval chunk prior to embedding, has been identified as a candidate improvement for reducing document-level retrieval mismatch, but has not yet been implemented in the current retrieval pipeline.

F. UI Redesign Completion

The interface redesign toward a lighter theme with a dark-mode toggle is in progress but not yet complete or deployed.

IX. Discussion

Table IV summarizes the completed-versus-pending status described above. The pattern that emerges is that the core technical pipeline – OCR, retrieval, grounded generation, and multilingual audio output – is functional, while the outstanding items are concentrated in two areas: user-facing convenience features (download, history) and evaluation rigor (quantitative metrics, documented fallback behavior). This is a common trajectory for a prototype moving from a working technical demonstration toward a deployable and defensible system, and it suggests a natural two-track roadmap: closing the user-facing gaps to make the tool practically usable by its target audience, and closing the evaluation gaps to make its claims of hallucination mitigation and retrieval quality verifiable rather than asserted.

The literature and product comparison indicate that the project's contribution is primarily an application-level combination rather than a new underlying AI technique. RAG for legal summarization is an active research area; however, the reviewed landscape leaves room for a system that combines document-grounded retrieval, multilingual text and audio output, OCR for scanned documents, and a free/open-source, low-resource deployment model for rural and low-literacy users. The analysis also shows that retrieval quality and legal correctness must be evaluated explicitly, because high text-overlap scores do not guarantee legally reliable summaries.

TABLE IV
Implementation Status Summary

Component	Status
Multilingual OCR (eng/hin/mar)	Completed
Curated knowledge base (5 domains)	Completed
Hybrid dense + BM25 retrieval	Completed
Grounded LLM summarization	Completed
Multilingual text output	Completed
Audio output (gTTS)	Completed
Session handling (no login)	Completed
Summary download / export	Pending
Previous-summary history view	Pending
Quantitative retrieval evaluation	Pending
Documented "not found" fallback	Pending
Summary-Augmented Chunking (SAC)	Pending
UI redesign (light theme, dark mode)	In progress

X. Conclusion

This paper has reviewed the academic and product landscape surrounding legal document summarization and access-to-justice technology in India, positioned the Rural Legal Aid Summarizer within that landscape, and documented its implementation status. The core RAG-based pipeline – multilingual OCR, hybrid retrieval, grounded summarization, and multilingual audio output – is complete and operational on a low-resource, free/open-source stack. The remaining work is concentrated in export and history features that affect day-to-day usability for the target rural user, and in the evaluation and documentation practices that would allow the project's hallucination-mitigation and retrieval-quality claims to be independently verified. Closing these gaps is the immediate development priority.

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